

Thus, an economic system must exist to determine the pattern of production of various goods and services required to satisfy the needs of the community. In other words, it must decide as to (1) which economic goods and services will be produced, (2) how will these be produced, and (3) in what quantities will these be produced.

In the modern economies all allocative decisions are made through the (i) market mechanism, and (ii) government decision. Alternatively, the market allocative mechanism may be designated as the private sector and the government allocation institution as the public sector. The market forces of demand and supply and the price mechanism, as determined by consumer sovereignty and producers' profit motives, characterise the private sector's allocation of scarce resources. The public sector's allocation, on the other hand, is accomplished through the revenue and expenditure activities of government budgeting. In reality, however, no society allocates its entire economic resources exclusively through a single allocative institution. Instead, each economy in the world is a "mixed" economy in the sense that it relies partly on the market-determined and partly on the government-determined resource allocation. Accordingly, an economy may typically be referred to either as a "capitalist" or a "socialist" economy depending upon the degree to which it stresses the market mechanism or government authority for the purpose of resource allocation. In a mixed economic system, besides directing the resource allocation function, both the private and public sectors also determine the performance of the other major branches of economic activity. These consist of the distribution of income, stabilisation and economic growth functions. Consequently, the present analysis will be concerned not only with the influence of the public sector on resource allocation, it will also study the impact of the public sector on these other economic areas.

OPTIMAL BUDGET

Before defining the concept of the optimal budget, the meaning of budget must be explained. A budget consists of the following three policies.

- (i) Revenue Policy,
- (ii) Expenditure Policy, and
- (iii) Debt Policy.

All these three policies are adopted in order to frame the budget. The concept of optimality is applied to all these three policies. The government should frame its budget in such a way that it is optimal. In other words, if the budgeted expenditure is more than the revenue, the government should use the tax policy and compensate the deficit financing and *vice versa*. When the overall effect of the budget on the economy of the country is the best, *i.e.*, when the economy gains the maximum social advantage through the

budget it is called an optimal effect. Richard Musgrave¹ has greatly emphasised the concept of optimal budgeting and he has suggested the following tests of the budget optimality:

1. **Maximum Rise in Per Capita National Income:** It is the most important test of the budget optimality. According to this criterion, the budget of the government must be framed in such a manner that it results in the optimum growth of per capita income in the country.

2. **Justice in the Distribution of Income:** According to this criterion of budget framing, the budget must aim at ensuring the distributive justice in the community. Consequently, the people should get income according to their capacity and work.

3. **Economic Stability:** According to this criterion, the budget should promote stability in the economic system.

Considering these three tests of the budget optimality, Musgrave has divided the entire optimal budget into the following three branches.

1. Allocation branch,
2. Distribution branch, and
3. Stabilisation branch.

John F. Due has, however, divided the optimal budget into following four parts.

1. Resource allocation,
2. Operational efficiency in resource uses,
3. Income redistribution, and
4. Economic stabilisation and growth.

The difference between Musgrave's approach and John F. Due's scheme the optimal budget analysis is only definitional. While Musgrave has included both the resource allocation and its operational efficiency under the allocation branch, John F. Due has studied these two separately. This difference notwithstanding, however, the main approach of both these economists is to achieve the maximum per capita income through the operations of optimal budgeting.

The fiscal department performs all these functions. The incharge of the budget allocation branch must determine the various adjustments that are required to be made in the allocation of resources, the people who will bear the cost, and the appropriate revenue and expenditure policies that will be required to achieve the desired objectives.

The distribution branch must determine the steps that are needed in order to establish the desired proper or just system that promises the distributive justice to people and the stabilisation branch must decide the measures that

¹ For a detailed discussion refer to Richard A. Musgrave, *The Theory of Public Finance—A Study in Public Economy*, McGraw-Hill Book Co., 1959.

are necessary to be taken in order to secure the price stability and full employment in the economy. In this way, the budget policy is determined on the basis of the three interdependent criteria each one of which involves the different objective and principles of action. These three criteria or tests are consolidated into a net budget involving a single set of tax and expenditure measures. The basic test of the optimal budget framing consists in an efficient budget planning according to each one of these three tests simultaneously.

1. Resource Allocation Branch

Resource allocation means the transfer of resources from the people to the government. It is done through taxation by reducing the disposable personal incomes of people in the country. The higher the level of taxation, higher would be the amount of total tax revenue transferred from private hands to public hands. While taxing the people, the government is faced with the formidable problem of simultaneously making the public policy as an effective instrument of securing a more efficient resource allocation. The question, however, is: why does government want to raise income and what part of the total goods and services produced in the economy should come under the cover or scope of the public sector and what part of these goods and services should be allowed in the private sector ?

The basic reason behind extending the scope of government activity is that the free market economy suffers from many evils and imperfections. The allocation policy must take into account the three important shortcomings of the market-oriented economy. Firstly, any optimal budget policy aiming at an efficient resource allocation must be aware of the existence of market imperfections and the budget policies must be directed to remove these imperfections which largely arise from the presence of private monopolies. However, the regulation of competition is not primarily a problem of budget policy. Generally, it is tackled through legislation seeking to control the industrial organisation and monopoly power, or to regulate the price and output policies of big firms.)

Secondly, the private sector does not want to bear large risks and uncertainties. The lumpiness of productive factors and production processes requires government intervention. Determination of optimal output requires the equating of average revenue with marginal revenue. Under the decreasing cost conditions, the firm cannot follow such a policy since it involves loss. A tax and subsidy process and hence budget policy is required to secure an optimal output.

Thirdly, social wants or collective consumption justifies government activity. If we assume that one person's welfare depends on the welfare of all others in the society then we must conclude that the satisfaction of all private wants involves gains and losses that are not accounted in the market.)

Public or Collective Goods (Certain goods and services are by nature such that they cannot be exchanged in the market. These can be made available to people only by the government.) For example, defence, police services, justice, bridge, highway, street-lighting, public parks, light-houses etc. are some examples of those services which cannot be made available through the free market pricing mechanism since their consumption is non-rival and those who are unwilling to pay the price for their consumption or enjoyment cannot be excluded from the consumption of such goods and services. These goods are known as public, social or collective goods. The following are the main characteristics of these goods.

1. They are jointly supplied to the community as a whole.
2. They are indivisible and provide service which is not divisible into units.
3. They are of collective benefit to the society.
4. The consumption of these goods by one person or group of persons does not reduce the amount of these goods available for consumption by the others in the community. In other words, their consumption is non-rival. For example, in the absence of overcrowding, any use of a river bridge, light-house, highway or public park by one individual does not in any way interfere with the similar use of these goods by others in the community.
5. It is not possible to exclude those people who do not pay a price from consuming these goods. Unlike the private goods whose consumption to a person can be denied if he does not pay the price for these goods, the consumption of public goods by such a person cannot be prevented through the free market mechanism. For example, if A pays for the use of street-lighting or traffic signals and if B does not pay, then B cannot be prevented from enjoying the benefit of such street-lighting and traffic signals since these are located in open public places and are, therefore, open for use by all the citizens.
6. These goods cannot be priced through the market mechanism. Consequently it creates the problem of raising finances required to produce public goods since few people would pay for the consumption of these goods voluntarily. This is known as the problem of *free-riders* implying that everyone would like to enjoy the consumption benefits of these goods without paying for the cost involved in their supply. Consequently, the state has to finance the cost of providing these goods through compulsory contributions by members of society in the form of taxes.
7. In real world economies, public goods are seldom found in their pure form in as much as seldom do these goods satisfy all the characteristics of pure public goods. So also very few goods could be called pure private goods. Generally, most goods are a mixture of 'publicness' and ' privateness' though in varying degrees. Such goods are called *quasi* or *impure* public goods. For example, in government schools, and hospitals education and

medical aid is free for the poor and their children, individuals whose income is above a certain prescribed minimum limit may be charged subsidised fees for the education of their children and medical aid for family.

According to Musgrave, public wants may be classified into (a) social wants, and (b) merit wants. Social wants are those wants of the community which are satisfied by government and the goods and services meant to satisfy these wants must be consumed in equal amounts by all. People who do not pay for these services cannot be excluded from enjoying the benefits of social wants. Such wants cannot be satisfied through the mechanism of the market because their enjoyment cannot be made subject to payment of market price.

Merit wants are those wants which are satisfied by the government with social wants. The government gives loans, grants etc. to the private sector in order to satisfy the merit wants. Such wants are met by those services that are subject to the exclusion principle and are satisfied by the market within the limit of effective demand. They become public wants if they are considered so meritorious that their satisfaction is provided through the public budget, over and above what is provided for through the market and is paid in the form of price by the private buyers. The provision for free education for children of the poor, supply of subsidised mid-day meals for poor children in schools because healthy and educated people are considered desirable for a progressive society, subsidised low cost housing and transport etc. are illustrations of merit wants.)

The satisfaction of merit wants cannot be explained in the same terms as the satisfaction of social wants. While both these are public wants in that both are provided for through the public budget, nevertheless the two categories of wants are subject to different principles.

Fourthly, the existence of the external economies and diseconomies compels the government to do much in the matter of resource allocation. (The term external economies and diseconomies refers to the *spill-over* or *neighbourhood* or *third party effects* the production of public goods. These effects may be felt in the form of economic benefits or losses to other economic units causing divergence between private and social marginal costs and benefits of the goods in question. For example, public transport system buses emitting poisonous smoke on roads in crowded areas in the city add to public health hazards increasing social cost of plying these buses.

Where the market mechanism involves varying degrees of inefficiency in resource allocation, the government has to play an important role in the economy requiring the transfer of large amount of resources from private to public sector.

The basic function of the allocation branch is to choose among the alternative rival uses of economic resources. Thus, the problem is essentially one of determining the opportunity cost. If resources are to be used for the

satisfaction of certain public wants, they will not be available for the satisfaction of private wants. In this basic sense, the budget of the allocation branch must always be balanced. The amount of resources withdrawn from the private use must equal the amount of resources added to the public use.)

2. Distribution Branch²

The distributional function (branch) of the optimal budget relates to the manner in which the "aggregate effective demand" for the economic goods is divided among the various individual and family spending units in the society. More specifically, this aggregate effective demand, which determines the distribution of the aggregate "real output" among the population, stems from the pattern of income and wealth distribution in the private sector and the pattern of political voting influence in the public sector. The distribution branch of the optimal budget transfers the income in the economy from one person to another. It also involves the concept of negative income-tax according to which those people who have income less than the minimum taxable income should be compensated or paid by the government. It is a device of transferring income from public sector to private sector.

The distribution of income and wealth in a free market economy depends on several factors, including the inheritance law, availability of educational opportunities, social mobility, and structure of markets. A mechanism must be provided by which appropriate required corrections can be made in the existing state of distribution of income and wealth in such a way that it does least damage to an efficient functioning of the economy. Such a mechanism is given by the tax and transfer system of the distribution branch. This is not, however, the only way in which adjustments in the state of distribution can be made. There are other ways also to bring about the required correction in the existing state of distribution of income and wealth in the economy. Minimum wage legislation, price support programmes for certain farm products, tariff protection, fair trade legislation, etc. are some of the policies that might be resorted to in order to bring about the much needed important distributional results. From the economic point of view, there is, however, an *a priori* preference for the budgetary approach.

To conclude, the distributional adjustments may be needed at times and such adjustments should be implemented through the distributional function of the optimal budget. However, the main difficulty is about deciding the

² According to some economists, the theory of just or equitable distribution lies outside the scope of economics. The issue has philosophical and political ramifications and should be left for the philosophers, poets and politicians to decide. However, the criteria of distributive justice, although have philosophical contents, are in practice constrained by economic considerations justifying economists' concern and interest in the theory of optimal distribution.

proper state of distribution. This decision cannot be made through the market process. For it, rather a political process of decision making is needed. Democratic thinking seems to establish a presumption in favour of equality, both political and economic, aiming at obtaining an equitable distribution of income in society. However, equity does not mean in equal parts but it involves the concept of distributing income justly. It really means the choice among the different interpretations which is a matter of value judgement and it is difficult to support or oppose a particular pattern of distribution of income in the community.

3. Economic Stabilisation and Growth

The economic stabilisation and growth function of the optimal budget is concerned with maintaining a high level of resource utilisation and a stable value of money unit. An uncontrolled free economy is characterised by fluctuations in prices and employment. Apart from relatively short term swings, secular maladjustments may create problems of unemployment and inflation in the economy. Public policy must, therefore, assume a stabilising function in order to limit the deviations from high employment and price stability within socially permissible limits. It is a situation where neither inflation nor depression exists in the economy and to achieve this objective compensatory financing scheme is important.

The basic logic of the compensatory finance is simple and its guiding principles may be summarised in the following words:

(a) **Full Employment:** If involuntary unemployment exists in the economy, then increase the level of aggregate demand so as to adjust the aggregate expenditure or outlay upward to the value of the aggregate real output produced at full employment.

(b) **Price Stability:** If inflation prevails in the economy, then reduce the level of the aggregate effective demand by adjusting the aggregate expenditure downward to the value of the aggregate real output measured in terms of the current prices rather than the rising prices.

(c) **Savings or Capital Formation and Investment:** If the aggregate saving exceeds the aggregate investment in the economy, then public investment outlay should be undertaken and *vice versa*. The aim should be to keep saving and investment in equilibrium in the economy at full employment level of aggregate output. For implementing the first two principles we need compensatory financing while for implementation of the third principle we need public investment. In other words, if full employment and price stability prevail in the economy, then maintain the level of the aggregate money expenditure in order to prevent the occurrence of unemployment and inflation.

Generally, there are two types of unemployment namely, the Keynesian and the structural. The Keynesian type of unemployment is due to the defi-

ciency of the aggregate effective demand in the economy. On the other hand, structural unemployment occurs due to changes in the structure of the economy and is not important when we talk about the economic stabilisation. According to the Keynesian technique of compensatory financing when there is deficiency of the aggregate effective demand in the economy, the government should increase its expenditure in order to compensate for a fall in the private expenditure in depression so that the level of the aggregate effective demand in the economy is maintained at a level high enough to permit full or near-full employment in the economy. On the other hand, when there is inflation or boom in the economy, the government should prune its expenditure to compensate for the unduly large increase in the private expenditure so that the level of the aggregate effective demand can be maintained at full employment. By applying this technique of public finance, we can achieve economic stabilisation at full employment in the economy. This technique is known as compensatory financing and is used in the stabilisation branch of the optimal budgeting.

Like the use of the technique of compensatory financing in order to achieve economic stability, we use the technique of development finance in order to promote growth in the economy. The technique of development finance is particularly used in the less developed countries. The following three things are important in the context of the development financing technique:

(a) **Increase in the Level of Capital Formation:** Since the per capita income, consumption and saving are low in the underdeveloped countries, capital formation is also at a low level because savings which feed capital formation are very low. Since people are steeped in stark poverty, capital formation cannot be brought about in required measure by the efforts of the private sector alone. Consequently, the public sector should be developed in order to promote capital formation to initiate a continuous process of capital formation and growth in the economy.

(b) **Use of New Technology:** Since the use of new techniques of production requires costly imported capital equipment which private enterprise may not be willing and able to import we have to depend on the public sector enterprise. Unless we use the new technology, the process of sustained economic development or growth in the underdeveloped countries is not possible.

(c) **Human Capital Formation:** Apart from promoting the physical or non-human capital formation, human capital formation should also be promoted. This requires massive investment in health, education, training and skills of people in the economy which can only be made by the public sector because apart from the heavy amount of investment which may be beyond the capacity of the private sector, such investment is also unattractive for the private sector on account of extremely low rate of return on such investment.

Fiscal adjustments are not the only means to secure economic stabilisation. Other approaches such as the monetary and debt policies, wage and price con-